```
BUILD-MAX-HEAP(A)

A.heapSize = A.length

for i = A.length//2 downTo 0
    MAX-HEAPIFY(A, i)
```

Iterate through the nodes of the tree and run MAX-HEAPIFY on each.

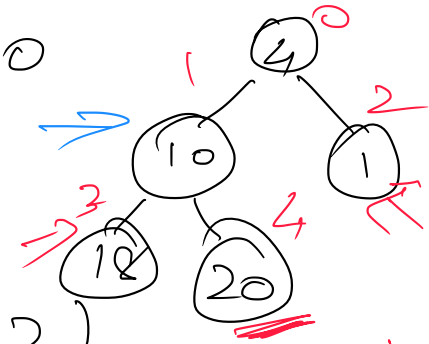
$A = [4, 10, 1, 18, 20]$ (indices 0 1 2 3 4)

$A.heapsize = 5$

$i = \frac{5}{2}$ to 0

$= 2$ to 0

$2, 1, 0$



First Iter: $i = 2$

Max-heapify(A, 2)

↳ $l = left(2) = NULL$

$r = right(2) = NULL$

nothing to heapify

Second Iter: $i = 1$

Max_heapify(A, 1)

↳ $l = left(1) = 2i+1 = 3$

$r = right(1) = 2i+2 = 4$

if $3 <= 5$ and $A[3] > A[1]$

(T) and $18 > 10$ (T) ✓

$largest = l = 3$

if $4 <= 5$ and $A[4] > A[3]$

(T) and $20 > 18$ (T) ✓

$largest = r = 4$

if $4 != 1$:

swap $A[1]$ w, $A[4]$

Nothing to heapify = Max_heapify(A, 4)

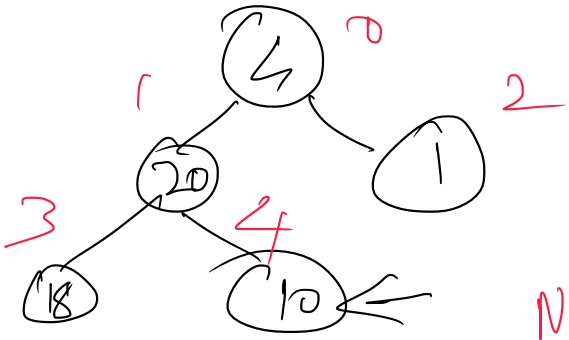
```
MAX-HEAPIFY(A, i)


  l = LEFT(i)
  r = RIGHT(i)

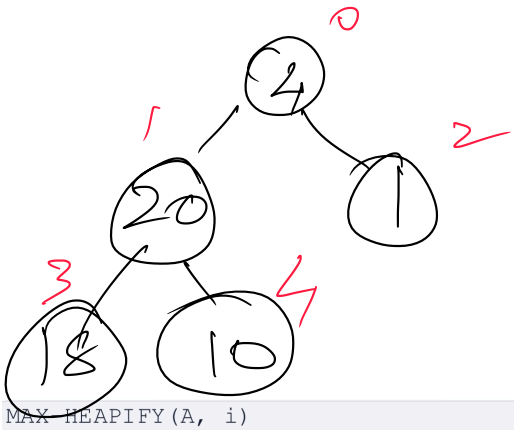

  if l <= A.heapSize and A[l] > A[i]
      largest = l
  else
      largest = i


  if r <= A.heapSize and A[r] > A[largest]
      largest = r


  if largest != i
      exchange A[i] with A[largest]
      MAX-HEAPIFY(A, largest)
```



Third Iter: $i = 0$



Max_heapify(A, 0)

↳ $l = 1$

$r = 2$

if $1 \leq 5$ and $A[1] > A[0]$

(T) $20 > 4$ ✓ (T)

largest = 1

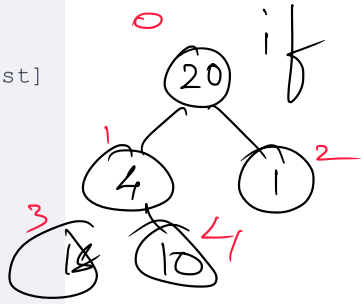
if $2 \leq 5$ and $A[2] > A[1]$

$1 > 20$ ✗

if $1 \neq 0$: ✓

exchange $A[0]$ w/ $A[1]$

max_heapify(A, 1)




```
MAX-HEAPIFY(A, i)

  l = LEFT(i)      = 2i+1
  r = RIGHT(i)     = 2i+2

  if l <= A.heapSize and A[l] > A[i]
      largest = l
  else
      largest = i

  if r <= A.heapSize and A[r] > A[largest]
      largest = r

  if largest != i
      exchange A[i] with A[largest]
      MAX-HEAPIFY(A, largest)
```

```
HEAPSORT(A)

  BUILD-MAX-HEAP(A)

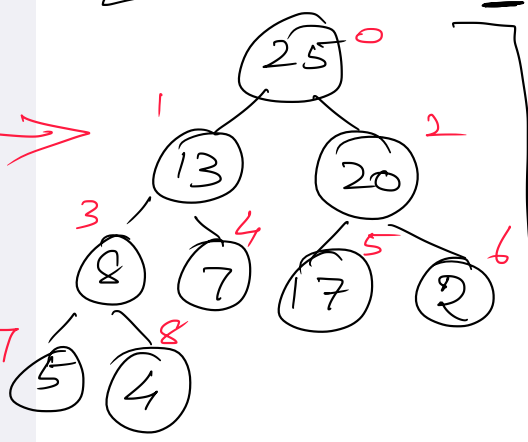
  for i = A.length-1 downTo 1

      exchange A[0] with A[i]

      A.heapSize = A.heapSize - 1

      MAX-HEAPIFY(A, 0)
```

$A = [25, 13, 20, 8, 7, 17, 2, 5, 4]$ (indices 0 1 2 3 4 5 6 7 8)



left = 2i+1

2·3+1

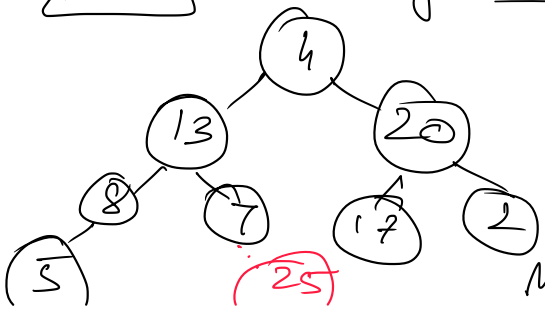
2·3+2

sort in ascending order

$i = 8$ downto 1

$i = 8$

exchange $A[0]$ w/ $A[8]$

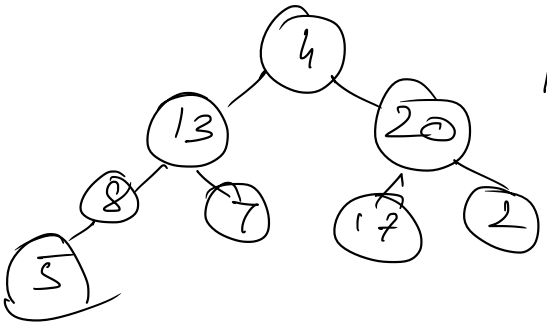


size = 8

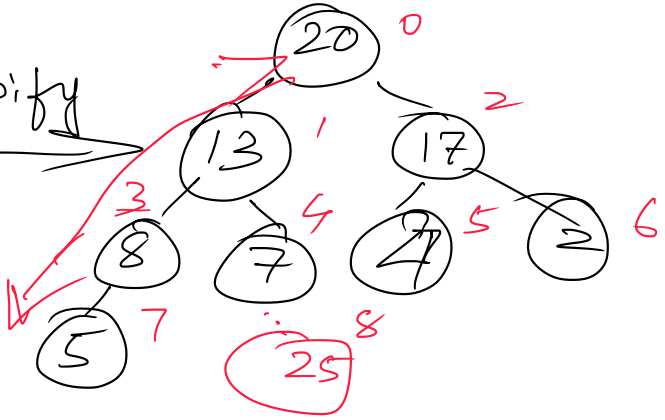
MAX_heapify(A, 0)

OUTPUT:





MaxHeapify

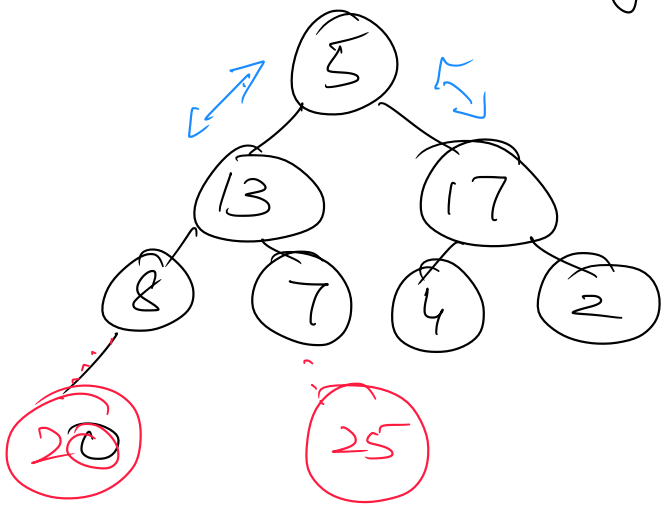


i = 7

exchange A[0] with A[7]

heapsize = 7

Max-heapify (A, 0)



i = 6

exchange A[0] w/ A[6]

